

PROJECT REPORT

FitBuddy

AI-Fitness Plan Generator

Using HTML, CSS and JavaScript

Project Type	AI / Web Application
Technology	HTML, CSS, JavaScript
Academic Level	2nd B.Sc. Computer Science with AI
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FitBuddy – AI Fitness Plan Generator

Project Development: 8 Phases

1. Brainstorming and Ideation Phase

The brainstorming and ideation phase is the first and most important stage in the development of the FitBuddy – AI Fitness Plan Generator project. The main purpose of this phase is to identify a real-world problem related to fitness and develop a suitable software-based solution. Many people are interested in maintaining a healthy lifestyle, but they may not know how to create a suitable workout routine or organize their fitness activities according to their personal goals. Beginners may find it difficult to decide which exercises to perform, how frequently they should exercise, and how to maintain a consistent routine. This problem provided the basic idea for developing FitBuddy.

During the brainstorming process, several ideas related to fitness applications were considered, including workout tracking, exercise recommendations, nutrition guidance, fitness reminders, progress monitoring, and personalized fitness plans. After evaluating these ideas, an AI-based fitness plan generator was selected as the main concept. The proposed system is designed to collect suitable information from the user and use it to generate a personalized fitness plan. The system can consider factors such as the user's fitness goal, activity level, available workout time, and preferred type of exercise.

The main objective of FitBuddy is to make fitness planning simple and accessible through a digital platform. Instead of searching through multiple websites or applications, a user can provide the required information in one place. The application can organize suitable workout recommendations and present them in a clear format. The project also aims to provide a simple interface so that users with different levels of technical knowledge can use the system comfortably.

The brainstorming stage also helps identify the major features required for the application. These include user profile creation, fitness goal selection, workout plan generation, general nutrition guidance, AI-based recommendations, and progress monitoring. The project can be developed using a modular structure so that additional features can be added later. At the end of this phase, the problem, proposed solution, target users, major features, and expected outcome are clearly identified. This idea becomes the foundation for requirement analysis, design, planning, development, testing, documentation, and demonstration.

Coding

```
<!DOCTYPE html>
<html><head><title>FitBuddy</title></head><body>
<h1>FitBuddy - AI Fitness Plan Generator</h1>
<p>Create a personalized fitness plan based on your fitness goal.</p>
<input id="name" placeholder="Enter your name">
<select id="goal">
<option value="">Select Fitness Goal</option>
<option>General Fitness</option><option>Strength Improvement</option>
<option>Muscle Building</option><option>Weight Management</option>
</select>
<button onclick="createPlan()">Create Fitness Plan</button>
<p id="result"></p>
<script>
function createPlan(){
let name=document.getElementById("name").value;
let goal=document.getElementById("goal").value;
if(name====" || goal===="){
document.getElementById("result").innerHTML="Please enter your name and select a goal.";
return;
}
```

2. Requirement Analysis Phase

The requirement analysis phase identifies and organizes everything the FitBuddy application must provide. After the project idea is finalized, the development team needs to understand user expectations, technical needs, system behavior, and limitations. Proper requirement analysis creates a clear foundation for the remaining stages and reduces errors during development.

The main functional requirement is that users should be able to enter their basic information and fitness preferences. The application may collect name, age, activity level, fitness goal, available workout time, and preferred exercise type. The system should validate the entered information before generating a plan. Users should also be able to view their generated workout recommendations in a structured manner.

Another major requirement is personalized plan generation. The system should identify the user's selected goal and generate relevant recommendations. The application can include workout routines, general nutrition guidance, and progress-related information. If artificial intelligence is integrated, the system should send relevant user inputs to the AI service and display the returned recommendations in a readable form.

Non-functional requirements include usability, reliability, responsiveness, performance, maintainability, and responsible handling of user information. The application should have a simple interface, provide quick responses, work on common devices, and handle incorrect or incomplete input properly. The system should avoid collecting unnecessary personal information and should clearly present fitness information as general guidance.

The software requirements can include HTML, CSS, JavaScript, a suitable backend technology, and an AI model or API. A modern web browser and a development environment are required for implementation and testing. At the end of this phase, functional requirements, non-functional requirements, hardware and software requirements, user needs, and system limitations are documented so that they can guide the design and development of FitBuddy.

Coding

```
<!DOCTYPE html>
<html><body>
<h2>FitBuddy - User Requirements</h2>
<form id="fitnessForm">
<input id="name" placeholder="Name" required><br><br>
<input id="age" type="number" min="13" placeholder="Age" required><br><br>
<select id="goal" required><option value="">Select Goal</option>
<option>General Fitness</option><option>Strength Improvement</option>
<option>Muscle Building</option><option>Weight Management</option></select><br><br>
<select id="activity" required><option value="">Activity Level</option>
<option>Beginner</option><option>Moderate</option><option>Active</option></select><br><br>
<select id="time" required><option value="">Workout Time</option>
<option>15 Minutes</option><option>30 Minutes</option>
<option>45 Minutes</option><option>60 Minutes</option></select><br><br>
<button type="submit">Submit Requirements</button>
</form>
<p id="message"></p>
<script>
document.getElementById("fitnessForm").addEventListener("submit",function(e){
  e.preventDefault();
  document.getElementById("message").innerHTML="Profile requirements successfully recorded.";
});
</script></body></html>
```

3. Project Design Phase

The project design phase converts the requirements into a structured model for the FitBuddy application. The purpose of this phase is to decide how the system will look, how users will move through it, how information will flow, and how individual modules will communicate. A clear design acts as a blueprint for the development stage.

FitBuddy can be divided into modules such as user profile management, fitness goal selection, workout generation, nutrition guidance, AI recommendation, and progress tracking. Each module performs a specific responsibility. The modular approach makes development easier and allows individual components to be tested separately before they are integrated.

The user interface design begins with a home page introducing FitBuddy and its purpose. The user can then move to a profile form where required information is entered. A goal-selection section allows the user to select an objective. After validation, the system processes the information and produces a fitness plan. A result page or result card can display the generated recommendations in an organized way.

The system workflow follows a simple sequence: user input, validation, processing, recommendation generation, and result display. If an AI service is used, the application can send the relevant information to a backend or AI endpoint and receive a response. Error messages should be displayed when information is incomplete or an external service cannot be reached.

The design also considers responsiveness and readability. Buttons, forms, headings, cards, and result sections should be arranged clearly and should remain usable on mobile and desktop screens. The final design should be simple enough for beginners while still providing enough information for users to understand their fitness plan.

Coding

```
<!DOCTYPE html>
<html><head><style>
body{font-family:Arial;background:#f4f6f8}.modules{display:grid;grid-template-columns:1fr 1fr;gap:18px;wid
</style></head><body>
<h1 style="text-align:center">FitBuddy Modules</h1>
<div class="modules">
<div class="module"><h3>User Profile</h3><button onclick="openModule('User Profile')">Open</button></div>
<div class="module"><h3>Fitness Goals</h3><button onclick="openModule('Fitness Goals')">Open</button></div>
<div class="module"><h3>Workout Plan</h3><button onclick="openModule('Workout Plan')">Open</button></div>
<div class="module"><h3>Nutrition</h3><button onclick="openModule('Nutrition')">Open</button></div>
</div>
<p id="status" style="text-align:center"></p>
<script>function openModule(x){document.getElementById("status").innerHTML="Selected: "+x;}</script>
</body></html>
```

4. Project Planning Phase

The project planning phase organizes the activities, resources, technologies, modules, and schedule required to complete FitBuddy successfully. Once requirements and design are finalized, the project is divided into manageable tasks so that development can proceed in a systematic manner. Planning also helps identify possible risks and establish methods for handling them.

The project scope is defined first. The core scope includes user information collection, fitness goal selection, personalized workout generation, general nutrition guidance, AI integration, and result presentation. Progress tracking, reminders, reports, and additional personalization can be considered depending on project requirements and available development time.

Technology planning is also performed in this stage. HTML can provide the structure of the web pages, CSS can create the visual design, and JavaScript can handle interactions, validation, and application logic. A backend can manage data and communicate with an AI service. An AI model or API can be integrated to produce personalized recommendations based on the user's input.

The project can be divided into modules and assigned a development sequence. First, the user interface and profile form can be implemented. Next, goal selection and validation can be added. Workout and nutrition modules can then be developed, followed by AI integration. After individual modules are completed, they can be integrated and tested as a complete application.

Risk planning is another important part of this phase. Possible issues include invalid user input, API failures, incomplete data, application errors, and delays in development. These risks can be reduced through validation, error handling, modular development, regular testing, backups, and proper documentation. A clear project plan therefore provides a roadmap from development through testing, documentation, and final demonstration.

Coding

```
const projectPlan={
  projectName:"FitBuddy - AI Fitness Plan Generator",
  technologies:["HTML","CSS","JavaScript","Backend","AI Integration"],
  modules:["User Profile","Fitness Goal","Workout Plan","Nutrition","AI Recommendation","Progress Tracking"],
  stages:["Design","Development","Testing","Documentation","Demonstration"]
};
console.log(projectPlan);
```

5. Project Development Phase

The project development phase is the main implementation stage of FitBuddy. In this phase, the ideas, requirements, design, and project plan are converted into a working software application. Development includes creating the frontend interface, implementing application logic, validating user input, generating fitness recommendations, connecting modules, and integrating AI functionality where required.

The frontend is developed to provide an easy interaction between the user and the application. Forms are created to collect relevant information, and CSS is used to make the interface organized and responsive. JavaScript handles user actions, validation, goal selection, and dynamic result generation. The application is structured so that the user can move from profile creation to goal selection and then to the generated fitness plan.

The fitness-plan module forms the central part of the application. Based on the user's selected goal and activity level, the system can prepare suitable workout suggestions. Different goals can have different recommendation sets. General fitness can focus on balanced activities, while strength-related goals can focus on strength-oriented movements. The system should keep recommendations organized by day or activity category.

The nutrition module can provide general healthy-eating guidance related to the user's selected goal. It should avoid presenting medical treatment or individualized medical advice. The AI component can be connected through a backend service or API. The user information can be formatted into a prompt, sent to the AI service, and the returned response can be displayed in the result section.

During development, each module should be implemented and checked individually before complete integration. This approach helps identify problems early and makes the code easier to maintain. At the end of development, the application should allow a user to enter information, select a goal, generate a plan, and view the result through one connected workflow.

Coding

```
function generatePlan(goal) {
  const plans = {
    "General Fitness": [
      "Warm-up",
      "Walking or light cardio",
      "Bodyweight exercises",
      "Stretching"
    ],
    "Strength Improvement": [
      "Warm-up",
      "Squats",
      "Push-ups",
      "Core exercises",
      "Stretching"
    ],
    "Muscle Building": [
      "Warm-up",
      "Resistance exercises",
      "Upper-body exercises",
      "Lower-body exercises",
      "Recovery"
    ],
    "Weight Management": [
      "Warm-up",
      "Brisk walking",
      "Bodyweight exercises",
      "Light cardio",
      "Stretching"
    ]
  };

  return plans[goal] || ["Select a valid fitness goal"];
```

6. Project Testing Phase

The project testing phase is carried out after the major components of FitBuddy have been developed. The main purpose of testing is to verify whether the application works according to the requirements and design. Testing helps identify errors, incorrect outputs, validation problems, interface issues, and integration failures before the final demonstration.

Unit testing can be performed on individual functions and modules. For example, the user-input validation function can be tested separately, and the fitness-plan generation function can be checked with different fitness goals. Each module should produce the expected result for valid inputs and handle invalid inputs appropriately.

Functional testing verifies whether the main features work correctly. The tester can enter user information, select different fitness goals, submit the form, generate a plan, and check whether the correct result is displayed. Different combinations of activity levels and workout durations can also be used to verify that the application responds consistently.

Input validation is particularly important because incomplete or incorrect information can affect the generated result. The system should identify empty fields, invalid age values, missing goals, and unsupported selections. Error messages should be clear and should guide the user to correct the input.

Integration testing checks whether different modules work together correctly. The profile module should communicate correctly with the goal-selection and plan-generation modules. If an AI API or backend is used, the connection should also be tested. The application should handle network or service failures without crashing.

User-interface testing checks whether buttons, forms, menus, result cards, and other elements work properly on different screen sizes. Performance and basic security checks should also be considered. After all identified issues are corrected, regression testing can be performed to make sure that changes have not damaged previously working features. The final testing stage ensures that FitBuddy is ready for documentation and demonstration.

Coding

```
function validateFitnessInput(data) {
  const errors = [];

  if (!data.name || data.name.trim() === "") {
    errors.push("Name is required");
  }

  if (!data.age || data.age < 13) {
    errors.push("Valid age is required");
  }

  if (!data.goal) {
    errors.push("Fitness goal is required");
  }

  if (!data.activity) {
    errors.push("Activity level is required");
  }

  return {
    valid: errors.length === 0,
    errors: errors
  };
}

const testData = {
  name: "User",
```

7. Project Documentation Phase

The project documentation phase records all important information related to the FitBuddy application. Documentation is necessary because it explains the purpose of the project, the problems addressed, system requirements, design, implementation, testing process, and final results. A well-prepared document also makes it easier for another person to understand, operate, maintain, or further develop the application.

The project report should begin with the project title, introduction, purpose, objectives, problem statement, and scope. The eight development phases can then be presented in sequence. Requirements, system design, technology selection, modules, development process, testing results, and final demonstration should be documented clearly.

Technical documentation can include the technologies used, project structure, important modules, input and output details, API or AI integration information, and installation or setup instructions. Code should be organized into appropriate files or modules so that it can be maintained easily. Screenshots of the user interface and generated results can be included to demonstrate the actual working system.

User documentation should explain how to open the application, enter information, select a fitness goal, generate a plan, and understand the displayed result. Instructions should use simple language and should not assume advanced technical knowledge. Any important limitations of the application should also be recorded.

Documentation should also contain testing details, including test cases and observed results. If a feature is planned for future development but is not currently implemented, it should be identified clearly rather than presented as completed. Proper documentation provides evidence of the development process and becomes an important part of the final project submission.

The final document can include the conclusion and future enhancement section. Possible future improvements include improved AI personalization, progress charts, reminders, additional exercise libraries, multilingual support, and integration with wearable devices. Complete documentation ensures that the FitBuddy project is clearly presented from its initial idea through its final implementation.

Coding

```
const documentation = {
  title: "FitBuddy - AI Fitness Plan Generator",
  sections: [
    "Introduction",
    "Problem Statement",
    "Requirements",
    "System Design",
    "Project Planning",
    "Development",
    "Testing",
    "Demonstration",
    "Conclusion",
    "Future Enhancements"
  ],
  technologies: [
    "HTML",
    "CSS",
    "JavaScript",
    "AI Integration"
  ]
};

console.log(JSON.stringify(documentation, null, 2));
```

8. Project Demonstration Phase

The project demonstration phase is the final stage of the FitBuddy – AI Fitness Plan Generator project. The purpose of this phase is to present the completed application and demonstrate how the different modules work together. The demonstration should show the complete workflow from opening the application to entering user information, selecting a fitness goal, generating a fitness plan, and viewing the final result.

The demonstration begins with a short introduction to the project. The presenter can explain the problem identified during the brainstorming phase and describe how FitBuddy provides a digital solution. The main objectives and features of the application can then be introduced so that the audience understands what the project is designed to accomplish.

The next step is to demonstrate the user interface. The presenter can show the home page, user profile form, fitness goal selection, and other available modules. Sample input can be entered into the form, and the application can demonstrate its validation process. The presenter should show how the system responds when required information is missing as well as when valid information is submitted.

The core part of the demonstration is the fitness-plan generation process. After entering the required information and selecting a goal, the presenter can generate the plan and display the resulting workout recommendations. If AI integration is implemented, the demonstration can show how the user's requirements are processed and how personalized recommendations are returned. General nutrition guidance and other available features can also be demonstrated.

The final part of the demonstration explains the testing results and project outcome. The presenter can mention that the application was checked using different inputs and that validation and error handling were included. The advantages and current limitations of the system can be discussed objectively. Future enhancements such as better personalization, progress visualization, reminders, and wearable-device integration can also be presented.

A successful demonstration should be simple, organized, and focused on the actual working features of FitBuddy. It provides an opportunity to show how the eight development phases contributed to the final application. The project concludes with the idea that AI and web technologies can be combined to create a convenient fitness-planning application while keeping the system easy to use and expandable for future development.

Coding

```
function demonstrateFitBuddy(user) {
  if (!user.name || !user.goal) {
    return "Please complete the required information.";
  }

  return {
    user: user.name,
    goal: user.goal,
    status: "Fitness plan generated successfully",
    nextStep: "View your personalized plan"
  };
}

const demoUser = {
  name: "Demo User",
  goal: "General Fitness"
};

console.log(demonstrateFitBuddy(demoUser));
```